

PRACTICAL – 8

MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements

1. Aim

To design a normalized MySQL database for a StudentHub application containing students, events, and registrations; create an Entity–Relationship (ER) model; implement primary and foreign key relationships; connect the database with PHP using PDO; and demonstrate secure database operations using prepared statements.

2. Problem Definition

Design the database for StudentHub with students, events and registrations. Create a PHP database connection using MySQL and implement secure database access. The database should be normalized, primary keys and foreign keys should be correctly defined, and database errors should be handled properly.

3. Key Questions / Analysis / Interpretation

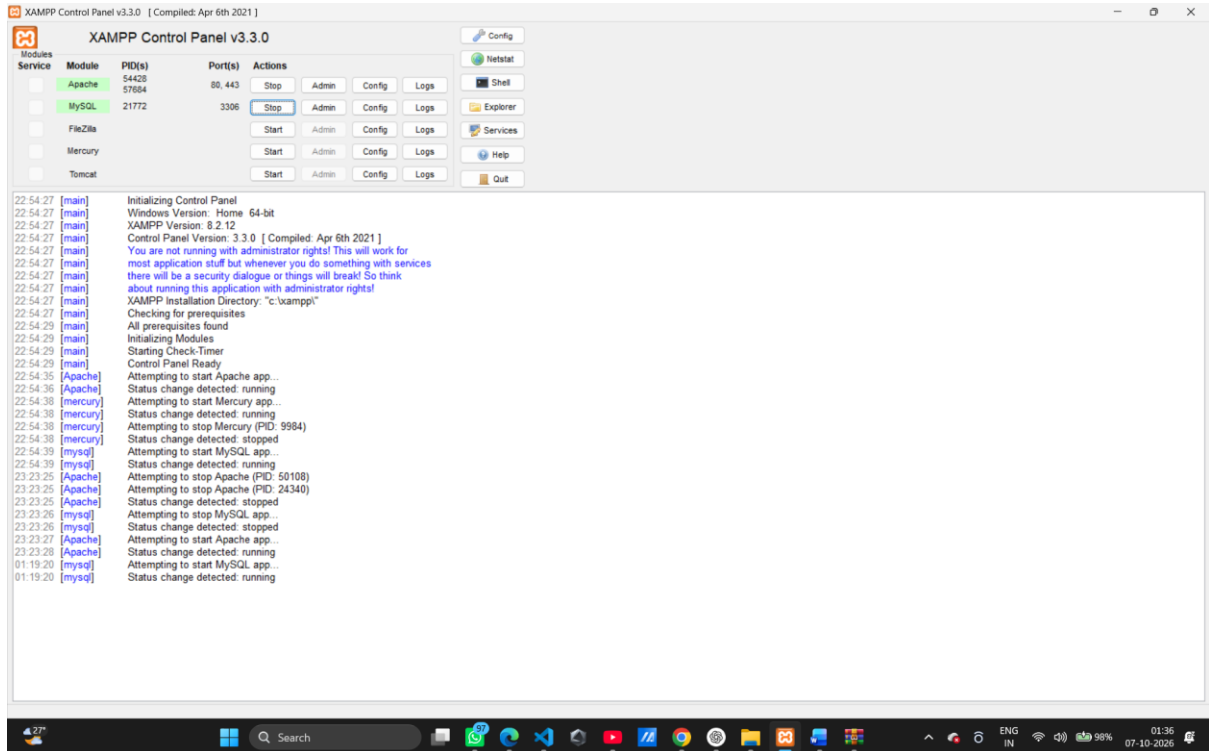
Question	Interpretation
1. Is the schema normalized properly?	Yes. Student, event, and registration information is separated into different tables, reducing duplicate data and update anomalies.
2. Are primary keys and foreign keys correctly defined?	Yes. Each table has a primary key. The registrations table contains foreign keys referencing students and events.
3. Is the database connection handled securely?	Yes. PHP PDO is used with exception handling and prepared statements. User input is passed as parameters instead of being directly concatenated into SQL.

4. Conclusion

The StudentHub database was designed using a normalized relational structure containing students, events, and registrations. Primary and foreign keys were used to maintain relationships and referential integrity. The database was connected to PHP using PDO, and prepared statements were used for secure INSERT and SELECT operations. The practical demonstrates the basic concepts required for building a secure database-backed web application.

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5. Output:



<http://localhost/phpmyadmin/index.php?route=/sql&pos=0&db=studenthub&table=students>

Server: 127.0.0.1
Databases
SQL
Status
User accounts
Export
Import
Settings

Databases

Create database

Create

☐ Check all

Database	Collation	Action
☐ information_schema	utf8_general_ci	Check privileges
☐ mysql	utf8mb4_general_ci	Check privileges
☐ performance_schema	utf8_general_ci	Check privileges
☐ phpmyadmin	utf8_bin	Check privileges
☐ studenthub	utf8mb4_general_ci	Check privileges
☐ test	latin1_swedish_ci	Check privileges

Total: 6

+
-
+
+
+
+
+

phpmyadmin
studenthub
New
events
registrations
students
test

←
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→

☐
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id
name
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course
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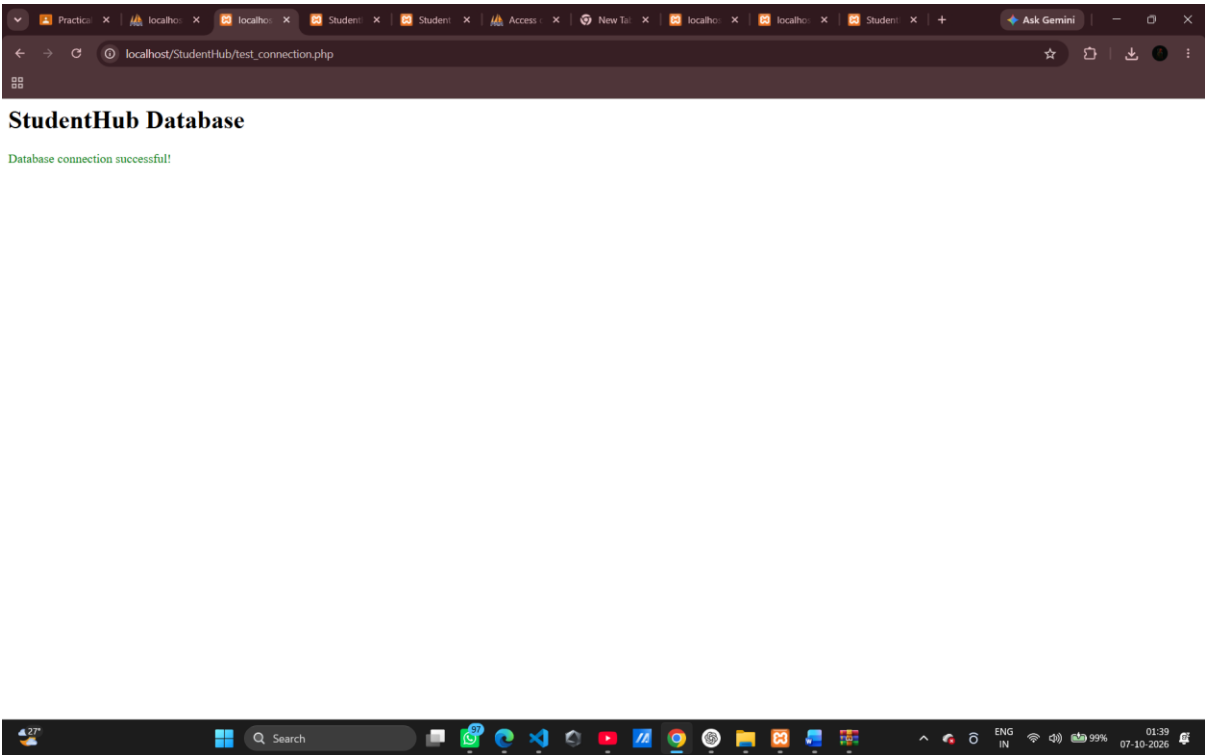
☐
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vatsal@gmail.com
Computer Engineering
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Rahul
rahul@gmail.com
Computer Engineering
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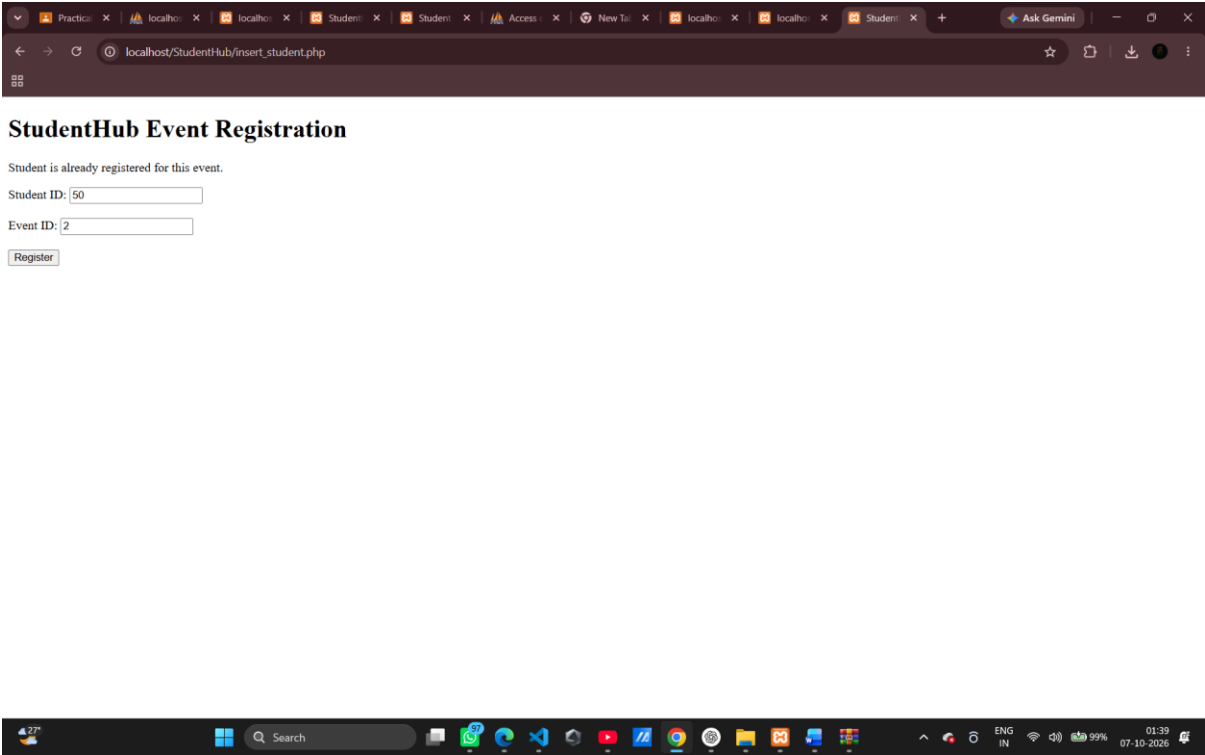
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Raj
raj@gmail.com
Information Technology
3

http://localhost/StudentHub/test_connection.php

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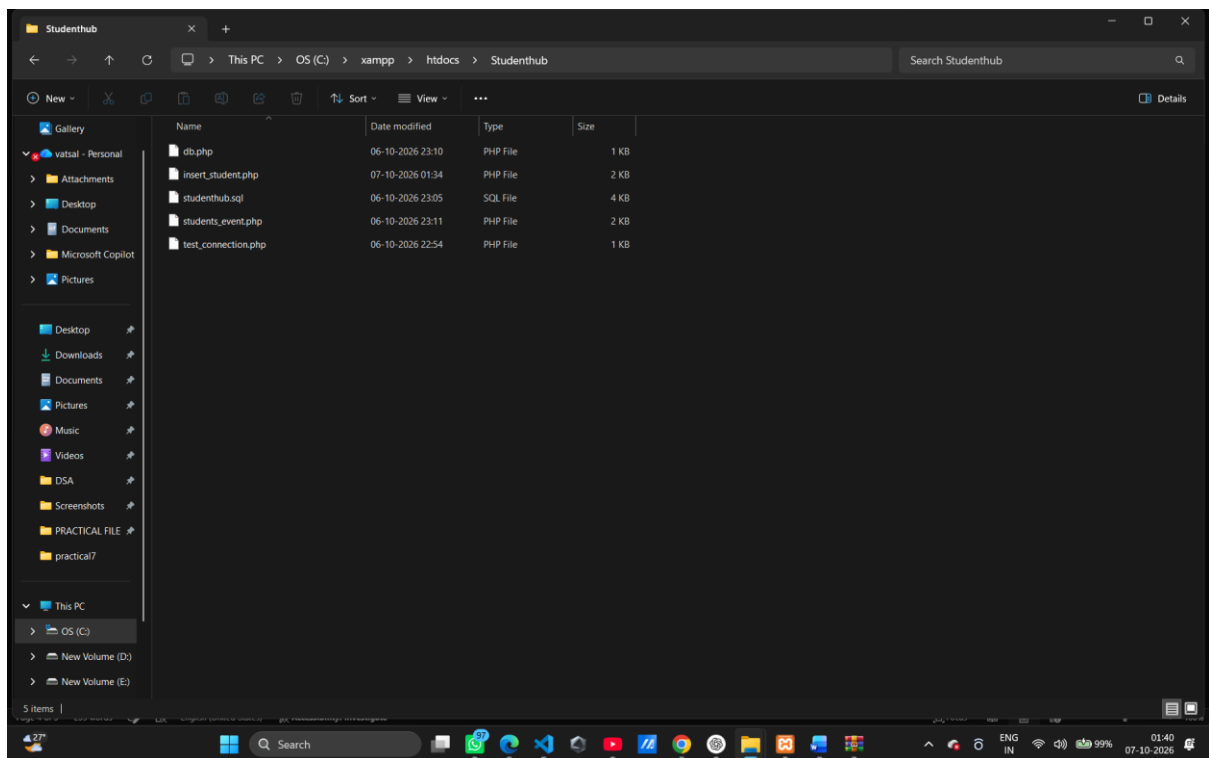
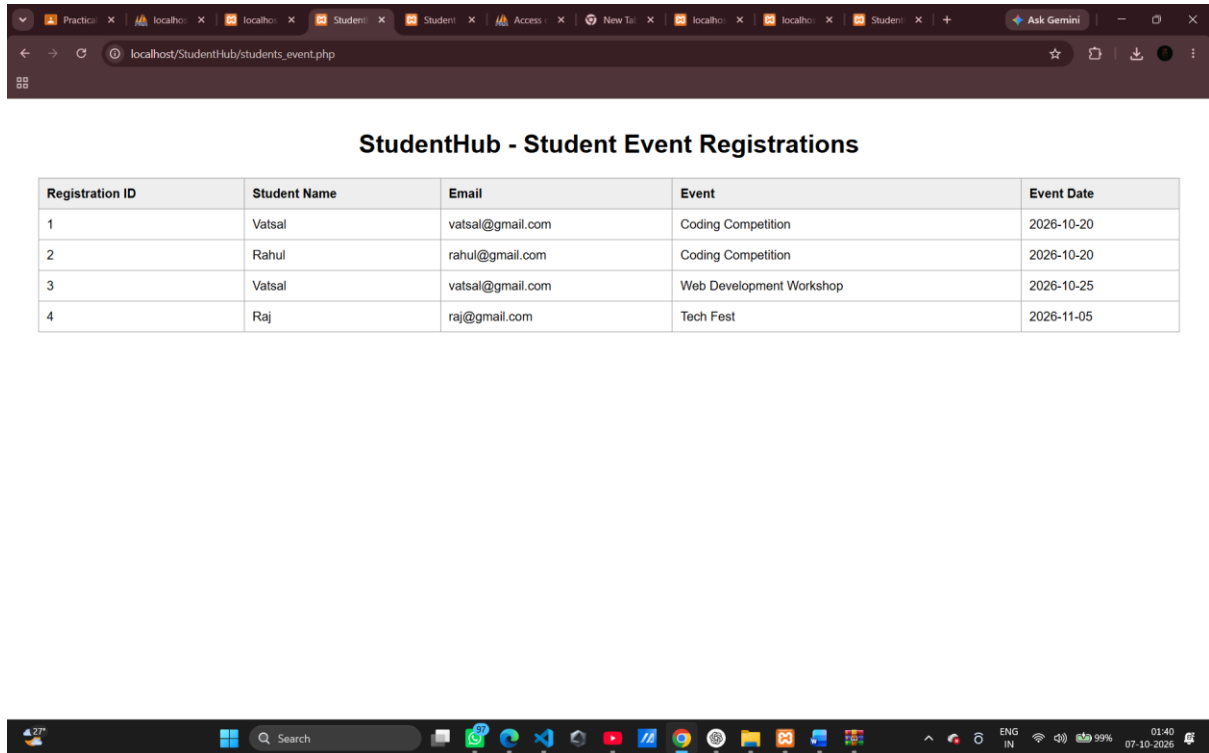


http://localhost/StudentHub/insert_student.php



http://localhost/StudentHub/students_event.php

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GitHub Repository Link:

<https://github.com/vatsal-1950/StudentHUB.git>